Though it might seems like you have very few pins when you start building applications, you'll soon find that most of your pins are still free!

RAM, EEPROM and Flash

- SRAM (Static RAM): Most Arduino™ boards have between 1 KB to 8 KB of SRAM, with the exception of Arduino™ Due that has 96 KB of SRAM.
- EEPROM (Electrically Erasable Programmable ROM): Arduino™ boards have EEPROM with capacity of 0.5 KB – 4 KB.
- Flash: Flash memory of Arduino™ boards is about 16 KB to 32 KB. Arduino™ Mega 2560 and Arduino™ Mega ADK have 256 KB Flash, while the Arduino™ Due has 512 KB Flash.

Memory not an issue

The size of your code file is completely different than the size of actual code that's flashed on the microcontroller. Unless you're using slightly complicated logic and many sensors, Flash memory of 16 KB and SRAM of 1 KB is enough.

Though these specifications might seem very low compared to the specs on your computer at home, they're enough to carry out most functions since the tasks these microcontrollers perform are fixed and don't need lots of memory. The Arduino™ programming environment does all the memory management tasks for your microcontroller.

So many options! What if I want to upgrade/change the board?

Many a time users think of changing boards, and moving from a smaller board to a bigger board is hardly an issue. But, there are problems when moving the other way around i.e. from a bigger board to a smaller board.

All Arduino™ boards are designed to be pin compatible with the original board. It simply means that any two Arduino™ boards have basic functionality pins at the same locations. Thus, any code that you've written for the original Arduino™ can be used directly for the new board with little or no modification. If the newer board has more pins like the Arduino™ Mega 2560, you'll see that the basic functionality pins remain the same. The only difference in these boards is that extra pins are added to interface more hardware.

If you're using the Analog Out functionality on the Arduino™ Due, this functionality is unavailable on any other boards, so you're stuck with the Arduino™ Due or might need to consider some newer boards.